Lab Report: Projectile Motion

Partner: Jamie | Period 3

Hypothesis

If a ball is launched horizontally from 1.2 m, the horizontal range depends only on initial velocity, since gravitational acceleration ($9.8\ m/s^2$) acts independently.

Data

Trial 1: $v_0 = 1.5$ m/s, range = 0.74 m

Trial 2: $v_0 = 2.0$ m/s, range = 0.98 m

Trial 3: $v_0 = 2.5$ m/s, range = 1.23 m

Theoretical range = $v_0 * \sqrt{2h/g} = v_0 * 0.495$

Analysis

Percent error across trials < 1.5%. Sources of error include air resistance and measurement precision. Newton's second law $F=ma$ and kinematic equations $v=u+at$, $s=ut+0.5at^2$ confirm results. Velocity, acceleration, displacement consistent with classical mechanics and conservation of energy.